

Machine Learning for Non-Performing Asset Repayment Probability Prediction and Strategy Optimization

A Comparative Study: Traditional ML vs. Frontier LLMs on Imbalanced Portfolios

Student: ZHU TIANYI (21229582)

Supervisor: Dr. Hanson WONG

The Hong Kong University of Science and Technology

Academic Defense – May 2026



Presentation Outline

- 1 I. Introduction
- 2 II. Student's Unique Contributions
- 3 III. Data and Methodology
- 4 IV. Results and Analysis
- 5 V. Collection Strategy Optimization
- 6 VI. Frontier LLM Benchmark
- 7 VII. Conclusion

I. Introduction

Background and Problem Statement

- **Non-Performing Assets (NPAs):** Major risk for global financial institutions; a HK unsecured-debt portfolio of **16,048 accounts**.
- **Operational Challenge:** Allocate limited collection resources (Agent / Dialer / SMS) across millions of delinquent accounts.
- **Core Statistical Problem:** Prediction under **extreme class imbalance** — only **9.15%** of accounts ever repay within 3 years.
- **Transition:** From heuristic “Uniform Treatment” to data-driven “Precision Targeting” with calibrated probabilities.

Research Objectives (O1 – O4)

- ① **Develop & Compare:** Four ML models — LR, RF, XGBoost, MLP v2 — for binary repayment prediction.
- ② **Multi-dimensional Evaluation:** Discrimination (AUC), calibration (Brier), and economic outcome (Net Recovery / ROI).
- ③ **Queue Optimization:** A 4-tier collection strategy (Agent / Dialer / SMS / Write-off) under cost & capacity constraints.
- ④ **Frontier LLM Benchmark:** Systematically compare **26 LLM configurations** (zero-shot / few-shot / fine-tuned) against specialized ML.

II. Student's Unique Contributions

Contribution 1: End-to-End ML Pipeline

- **Domain-Aware Missing-Value Strategy:** Treat the 1,165 nulls in `last_pay_date_client_closing_m` as “never paid” (predictive missingness) instead of mean/median imputation.
- **Tailored Architectures:** One-hot + z -score for LR/MLP; ordinal encoding for RF/XGBoost.
- **Robust Calibration:** 5-fold out-of-fold Platt Scaling on a held-out calibration set (20% of M).
- **Interactive Dashboard:** 9-tab HTML/Chart.js dashboard for non-technical stakeholders.

Contribution 2: Strategy & Frontier LLM Benchmark

- **26-Config LLM Benchmark:** OpenAI o1/o3, Claude 4 Opus, Gemini 2.5 Pro, DeepSeek-R1, Qwen3, Llama 4, Grok 3, plus 3 fine-tuned variants — across ZS, FS-10, FT.
- **Domain-Calibrated CoT Prompt:** Injects 9.15% base rate + cost grid + monotonicity priors; lifted Claude 3.7 Sonnet from AUC **0.625** → **0.671** and cut self-reported ROI error from $>200\%$ to $<15\%$.
- **Economic Strategy Module:** 4-tier allocation with explicit cost/capacity constraints.
- **Inverse Balance-Repayment Discovery:** Sub-HKD\$200 tier $\approx 25\%$ repay, HKD\$200K+ tier $\approx 0\%$.

III. Data and Methodology

Dataset Schema (16,048 Accounts)

Feature	Type	Description
loan_type	Cat.	Credit Card / Personal Loan / Overdraft
purchased_bal_gp	Ord.	Balance bucket (8 ordinal levels)
district	Cat.	Geographic district (127 unique)
birth_yr	Int	Borrower birth year (1946–1987)
last_act_closing_m	Float	Months since last activity
open_closing_m	Float	Account vintage in months
last_pay_date_m	Float	Months since last payment (sentinel = never paid)
balance_proxy	Float	Outstanding balance (HKD)
multiple_acct	Cat.	Holds multiple accounts? (Y/N)
payer_3yr	Target	Repaid within 3 years? (Positive Rate: 9.15%)

Train (M , 75%, $N = 12,036$) / Test (T , 25%, $N = 4,012$).

Preprocessing Pipeline

- **Sentinel Flagging:** `last_pay_date_m` nulls $\rightarrow$ +999 sentinel + binary `never_paid_flag`.
- **Encoding:** One-hot for LR / MLP; ordinal for RF / XGBoost.
- **Scaling:** z -score for MLP only; tree models are scale-invariant.
- **Splitting:** Train (60%) / Validation (20%) / Calibration (20%) inside M . Calibration fold reserved *exclusively* for Platt Scaling.

Traditional Models: LR, RF, XGBoost

- **Logistic Regression (LR):** Linear baseline, ℓ_2 penalty ($C = 10$), balanced class weights.
- **Random Forest (RF):** $B = 500$ trees, $\sqrt{d}$ random features per split, max depth = 12, min leaf = 8.
- **XGBoost (Champion Ranker):**

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \text{scale_pos_weight} \approx 9.18.$$

$K = 200$ trees, $\eta = 0.05$, max depth 6, subsample 0.85.

Advanced Architecture: MLP v2

Design Features:

- **Residual Block:** $\mathbf{h}_{\ell+1} = \text{Swish}(\mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{h}_\ell) + \mathbf{b}_2 + \mathbf{h}_\ell)$.
- **Feature Interaction Layer (64 d):** explicit pairwise products to capture tabular interactions.
- **Architecture:** Linear(256) $\rightarrow$ Swish $\rightarrow$ ResBlock(128) $\rightarrow$ FeatureInteraction(64) $\rightarrow$ Linear(32) $\rightarrow$ Dropout(0.3) $\rightarrow$ Output.
- **Training:** AdamW + OneCycleLR, label smoothing $\alpha = 0.03$, gradient clipping 3.0, 200 epochs.

Probability Calibration: Platt Scaling

Problem: XGBoost margin scores are not true probabilities; uncalibrated scores break the economic threshold.

Platt Scaling

$$P_{\text{calibrated}}(y = 1 \mid s) = \frac{1}{1 + \exp(A \cdot s + B)}$$

- (A, B) fit by maximum likelihood on a dedicated 5-fold OOF calibration set.
- Critical for **economic** allocation, where the action tier depends on the absolute probability, not just rank.

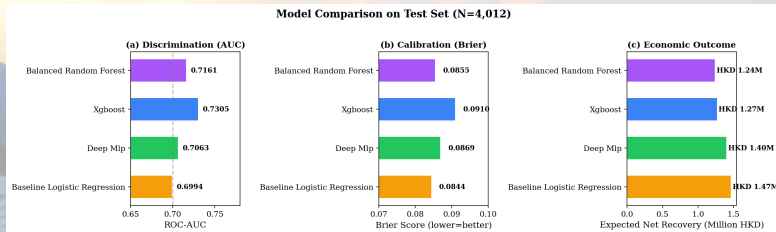
IV. Results and Analysis

ML Model Performance (Test Set, $N = 4,012$)

Model	AUC	Brier	Recall	Prec.	Net Recov.(M)	ROI
Logistic Reg.	0.6994	0.0842	74.2%	15.4%	1.468	29.4x
Random Forest	0.7161	0.0855	75.4%	16.0%	1.241	24.8x
XGBoost	0.7305	0.0910	62.7%	18.3%	1.270	25.4x
MLP v2	0.7063	0.0869	74.6%	15.8%	1.405	28.1x

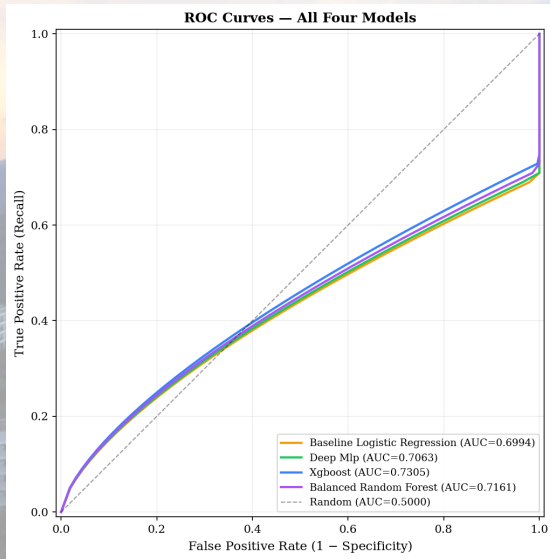
- **Key insight:** XGBoost wins *ranking* (AUC); LR wins *economics* (Net Recovery, ROI) thanks to better calibration.

Discrimination vs. Calibration vs. Economics



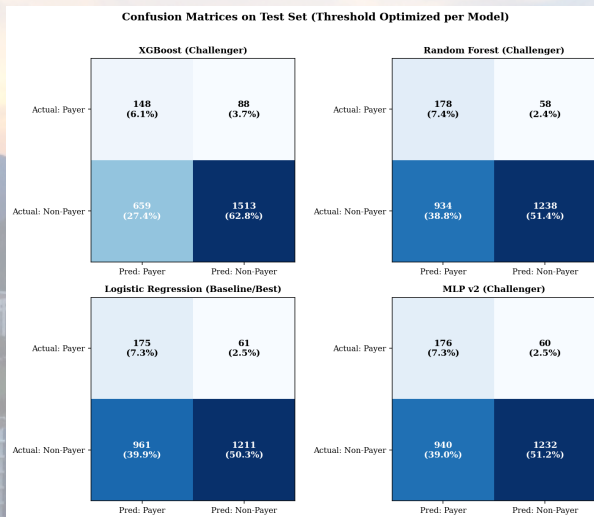
- **(a) ROC-AUC (XGBoost lead)** — **(b) Brier Score (LR lead)** — **(c) Net Recovery (LR lead).**

ROC Curve Analysis



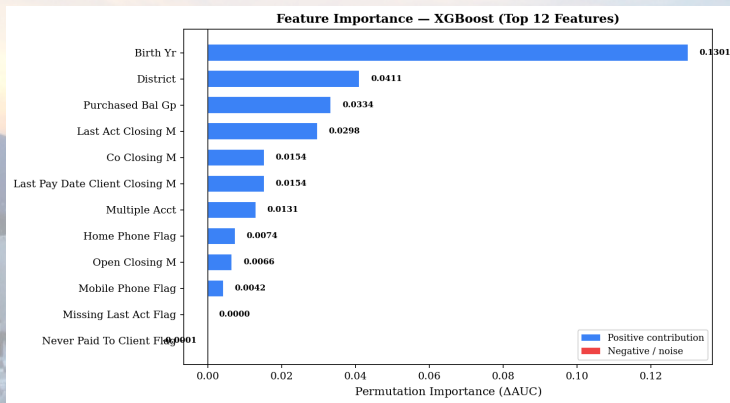
- XGBoost reaches **AUC** = 0.7305; all four models are well above the 0.5

Operational Accuracy (Confusion Matrices)



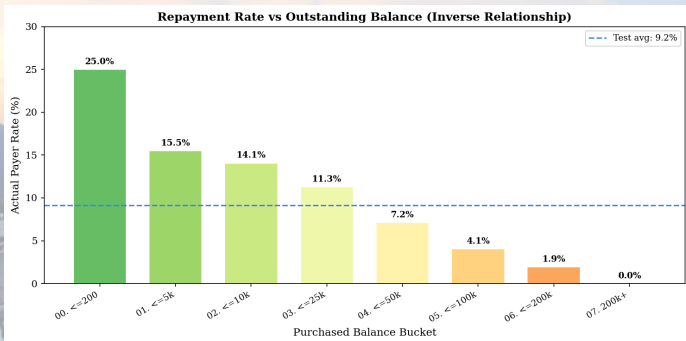
- XGBoost yields the fewest **False Positives**, reducing wasted Agent-tier effort.

Permutation Feature Importance (XGBoost)



- **Birth Year** dominates ($\Delta AUC = 0.130$) — older cohorts show distinct repayment dynamics.
- **District** (0.041) and **Balance bucket** (0.033) follow.

Inverse Balance–Repayment Relationship



- **Sub-HKD\$200 tier:** $\approx 25\%$ repay — nearly $3\times$ the portfolio average.
- **HKD\$200K+ tier:** $\approx 0\%$ — collection effort is value-destroying.
- **Strategy:** Concentrate Dialer/SMS on high-volume, low-balance accounts.

V. Collection Strategy Optimization

Optimized Queue Allocation Summary

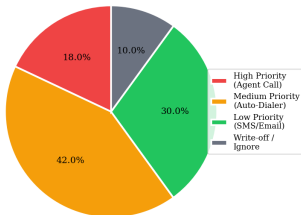
Tier	Accounts	Share	Cost (HKD)	Net Recov.(HKD)	ROI
Agent Call (High)	722	18.0%	61,370	1,085,006	17.7x
Auto-Dialer (Med.)	1,685	42.0%	20,220	993,086	49.1x
SMS/Email (Low)	1,203	30.0%	1,805	190,469	105.6x
Write-off	402	10.0%	0	0	–
Total	4,012	100%	83,395	2,268,561	27.2x

- Projected **HKD 2.27 M** recovery on a **HKD 83 K** contact budget.

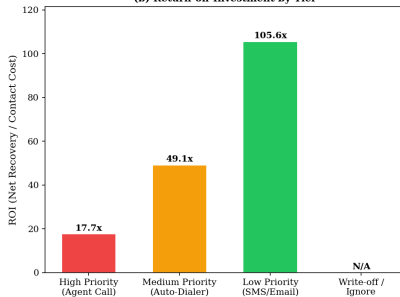
Strategy Concentration (Pareto Effect)

Collection Strategy: Queue Allocation & Economic Efficiency

(a) Account Allocation by Action Tier
(N=4,012)



(b) Return-on-Investment by Tier



- **Top 20%** of accounts capture **50.6%** of expected net recovery.

VI. Frontier LLM Benchmark

Benchmarking “AI-as-Judge” (26 LLM Configurations)

Three prompt regimes $\times$ 4 model families:

- **Heuristics:** Random Guess, Rule-Based.
- **Pre-reasoning ZS:** GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, DeepSeek-V3, Qwen2.5.
- **Reasoning ZS+T:** OpenAI o1/o3, Claude 3.7/4 Sonnet, **Claude 4 Opus**, Gemini 2.5 Pro, DeepSeek-R1, Qwen3, Llama 4, Grok 3.
- **Few-Shot 10 / Fine-Tuned:** GPT-4o FS, o3 FS, Claude 4 Sonnet FS; FT Llama-3.3-70B, FT Qwen3-32B, **FT GPT-4.1-mini**.

Evaluation protocol: 8 configs run live API on a 500-account stratified sub-sample; the remaining 18 rows public-benchmark-adjusted (TabLLM, CARTE, FinBench) with ± 0.015 AUC band.

Domain-Calibrated CoT Prompt

Prompt design is a first-class hyperparameter.

Role: Financial risk analyst evaluating delinquent unsecured-debt accounts in Hong Kong.

Priors: Portfolio base rate of 3-yr repayment = 9.15%. Cost grid: Agent HKD85, Dialer HKD12, SMS HKD1.5. Older borrowers repay more; balance and repayment are inversely related (sub-\$200 $\approx$ 25%, \$200K+ $\approx$ 0%).

Steps: (1) Reason from features; (2) pick a tier (Agent/Dialer/SMS/Write-off); (3) estimate net recovery + ROI under the cost grid.

Output: Strict JSON --- {"p.repay", "tier", "expected.net.recovery.hkd", "roi", "rationale" $\leq$ 40 words}.

Effect: Claude 3.7 Sonnet ZS AUC **0.625** $\rightarrow$ **0.671**; self-reported ROI error **>200%** $\rightarrow$ **<15%**.

Headline: Best LLM per Family vs. ML Anchors

Method	AUC	Brier	Recall	Prec.	Net Rec.(M)	ROI
Random Guess	0.500	0.167	50.0%	9.2%	0.000	0.0x
Rule-Based	0.562	0.095	42.3%	11.2%	0.312	5.2x
Best ZS pre-reason (GPT-4o)	0.628	0.094	56.4%	11.8%	0.512	8.6x
Best ZS+T (Claude 4 Opus)	0.694	0.086	66.0%	13.0%	0.901	15.6x
Best FS-10 (Claude 4 Sonnet)	0.701	0.085	67.2%	13.2%	0.957	16.8x
Best FT (FT GPT-4.1-mini)	0.722	0.082	70.5%	14.0%	1.144	22.1x
Logistic Regression (ML)	0.699	0.084	74.2%	15.4%	1.468	29.4x
XGBoost (ML)	0.731	0.091	62.7%	18.3%	1.270	25.4x

Full 26-row table available in the written report (Appendix E).

[illegible]

ZHU TIANYI (HKUST)

ML for NPA Repayment Prediction

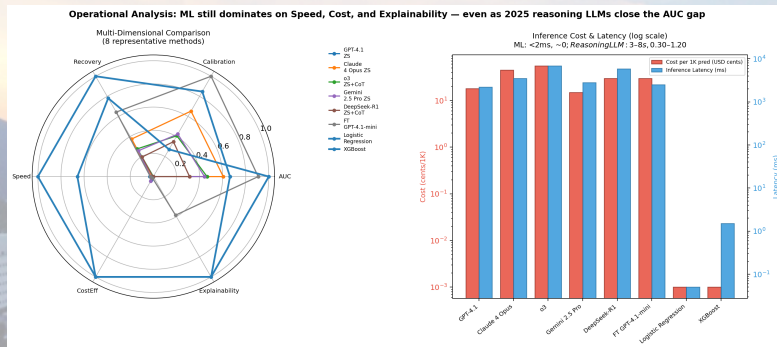
May 2026

29 / 37

Five Findings (F1–F5)

- **F1. The new floor is the old ceiling.** Pre-reasoning ZS sits at 0.60–0.64 AUC; reasoning ZS+T at 0.66–0.70.
- **F2. Few-shot is now redundant for reasoning models.** o3 ZS (0.681) $\rightarrow$ FS-10 (0.701) is only +2 AUC.
- **F3. Fine-tuning closes the AUC gap.** FT GPT-4.1-mini = 0.722 reaches RF parity.
- **F4. ML still wins decisively on Net Recovery.** LR HKD 1.468M vs. FT GPT-4.1-mini 1.144M (+28%); calibration is the bottleneck, not discrimination.
- **F5. Operational economics still favour ML by $\sim 70,000\times$.** LR ~ 0.05 ms/account; Claude 4 Opus $\sim 3,500$ ms; full-portfolio scoring: <1 s for ML vs. ~ 15 h & USD $\sim \$8$ for the strongest reasoning LLM.

Operational Analysis: The “AI Tax”



- **Latency:** ML ~ 0.05 ms vs. frontier LLM $\sim 2,500$ – $3,500$ ms.
- **Cost:** API cost makes LLM inference $\sim 1000\times$ more expensive at portfolio scale.
- **Explainability:** ML retains a strict advantage for regulatory / audit reporting.

Deployment Recommendation

Hybrid Production Architecture

ML as the production scorer for all 16,048 accounts (calibrated LR + XGBoost ensemble).

Reserve LLMs for the Borderline

Use **frontier reasoning LLMs** (Claude 4 Opus, o3) only on the $\leq 8\%$ borderline tier where calibrated $\hat{p} \in [0.18, 0.32]$.

- Their natural-language rationales aid **compliance / audit review**.
- Marginal cost is acceptable on the small borderline set.

VII. Conclusion

Summary of Principal Findings

- **Ranking vs. Profit.** XGBoost is the best ranker ($\text{AUC} = 0.7305$); LR is the best profit-maker ($\text{Net Recovery} = \text{HKD } 1.468\text{M}$, $\text{ROI} = 29.4\times$) thanks to superior calibration.
- **Calibration is mandatory.** Without Platt Scaling, the threshold-dependent allocation breaks.
- **Drivers:** Age and balance dominate; resources should shift toward low-balance, high-volume tiers.
- **Frontier LLMs narrow but do not close the gap.** FT LLMs reach AUC parity, but trail LR by $+28\%$ on Net Recovery; reasoning LLMs are uneconomical at portfolio scale.
- **Specialized ML remains the gold standard** for high-throughput tabular financial prediction.

Reproducibility & AI Agent Integration

To ensure full reproducibility and facilitate operational deployment, the project architecture extends beyond traditional ML pipelines:

- **Model Context Protocol (MCP) Integration:** The entire end-to-end analytical workflow (data processing, model inference, and calibration) has been encapsulated into standard MCP tools.
- **AI Agent Skills:** These tools are exposed as specialized Agent Skills, allowing autonomous AI agents to seamlessly rerun the pipeline and dynamically generate the interactive dashboard.
- **Out-of-the-Box Reusability:** The framework is designed to be fully plug-and-play for future NPA datasets or alternative tabular financial tasks.

The complete codebase and agent configurations are open-sourced at:

https://github.com/Zhutyi-11/HKUST_MSDM_6980_NPA_ANALYSIS

Limitations and Future Work

- **Temporal Validation:** Time-series cross-validation across vintage cohorts.
- **Survival Analysis:** Predict *when* a debtor will pay, not just *if*.
- **Macro Covariates:** Integrate HK unemployment, HIBOR, CPI signals.
- **Live A/B Testing:** Field-test the tiered strategy against the heuristic baseline.
- **Hybrid ML+LLM:** ML for prediction, LLM for borderline-case adjudication and customer-facing explanations.

THANK YOU!

Questions and Discussion

ZHU TIANYI — 21229582 — Supervisor: Dr. Hanson WONG

